

Problem-Solution Fit

Date	10 july 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

This canvas explains how the Emotion-Aware Learning Support Engine addresses the needs of students and educators by identifying emotional learning barriers, understanding their root causes, and providing an AI-powered ensemble solution for empathetic guidance.

1. Customer Segment (CS) Students, Educators, and Digital learning environments.	6. Customer Limitations (CL) Educators cannot gauge emotions; low personalization.	5. Available Solutions (AS) Generic AI tutors, LMS without analytics, static quizzes.
2. Problems / Pains (PR) Uniform responses regardless of student emotional state.	9. Problem Root Cause (RC) Text-only assistants lack emotional context parsing.	7. Behavior (BE) Students repeatedly re-read material or disengage.
3. Triggers to Act (TR) Student expresses frustration or confusion. 4. Emotions (EM) Before: Confused, anxious. After: Confident, supported.	10. Your Solution (SL) AI application using BiLSTM+DistilBERT to classify emotions and Gemini AI for empathetic guidance.	8. Channels of Behavior (CH) Online: Streamlit Web Application, API, Dashboards.